TRIANGLE

Counting Triangles, Squares and Rectangles in Figures

OAVS TGT | Reasoning (Non-Verbal) | Topic 12: third non-verbal topic

Syllabus line (Reasoning, 30 Marks): “The Non-Verbal Reasoning topics like Figure Formation, Dice, **Triangle**, Rule Detection, Images, Completion of Pattern, Mirror Images, Figure Matrix and Paper Folding, etc.”

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). **Only one counting-figure question appears in these papers** (English 2022 Q32, counting squares). The syllabus names ‘Triangle’, so this PDF teaches triangle counting as the expected sub-topic and uses the PYQ for the square-counting method. Every count in this PDF was verified by a counting program.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	The one PYQ and what it means for preparation
2	Foundation	Counting principles and the ‘no repeat’ routine
3	Counting Triangles	Apex lines, parallel lines, triangular grids, medians, diagonals; formulas and tables
4	Counting Squares and Rectangles	Grids, diagonals, and the PYQ solved step by step
5	Shortcuts, Traps and Scope	Speed methods and mistakes
6	Practice Set	12 original counting figures with a verified key
7	Quick Revision	One-page formula sheet

CHAPTER 1

PYQ Analysis

One counting question in five papers

Paper	Q. numbers	Count	What was asked
English 2019	none	0	—
English 2022	32	1	How many squares are there in the figure (a 4 × 4 grid with diagonals)
Mathematics 2024	none	0	—
Science 2024	none	0	—
Social Studies 2023	none	0	—
Total		**1**	**Answer 35**

PYQ INSIGHT: **No triangle-counting question appeared in these five papers.** The single counting question was about **squares** in a grid with diagonals, and it needed more than the 4 × 4 square formula (30 squares): the diagonals create 5 extra tilted squares, giving 35. Other papers of this exam family commonly ask for triangles, so triangle counting is included here as an expected sub-topic.

Pattern	PYQs	Priority
Counting squares in a grid with diagonals (tilted squares)	1	★★★ Must
Expected: counting triangles (apex lines, grids, medians, diagonals)	0	★★ Expected
Expected: rectangles in a grid	0	★★ Expected

SHORTCUT: **Preparation plan:** learn the formulas in Chapters 3 and 4 (about 10 minutes), then practise with figures. Counting questions are quick if you count by **size** and never by guess.

CHAPTER 2

Foundation

Counting principles

Principle	Meaning
A triangle needs three sides	Each side must lie along drawn lines, and the three lines must not meet at one point.
A triangle may contain many regions	Count single-region triangles, then two-region, three-region and so on, or use a formula.
A square needs four equal sides and four right angles	Squares can be tilted: sides along the diagonals count.
A rectangle counts squares too	'How many rectangles' includes all squares; 'how many squares' counts only squares.
No repeats, no misses	Count by size (smallest first) or by position (left to right).

Routine

Step	Do this
1	Label the corners and intersection points with letters.
2	Decide the figure type: apex lines, parallel lines, triangular grid, grid of squares, diagonals.
3	Use the formula for the type (see the tables) or count by size.
4	Check one hand-counted size against the formula before trusting it.

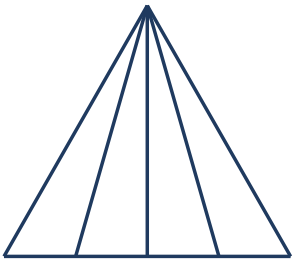
CHAPTER 3

Counting Triangles

Formulas and verified examples

3.1 Lines from the apex to the base

If the base is cut into **k** parts by lines from the apex (k − 1 inner lines), the triangles are formed by choosing two of the k + 1 lines from the apex: **k (k + 1) ÷ 2**.

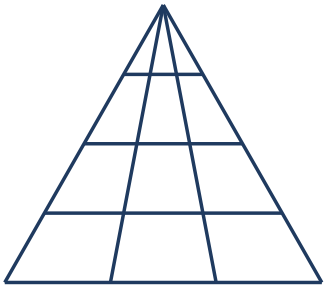


Example: 3 inner lines, so k = 4. 4 × 5 ÷ 2 = **10** triangles. (Program count: 10.)

k	1	2	3	4	5	6	7	8
Triangles	1	3	6	10	15	21	28	36

3.2 Apex lines and lines parallel to the base

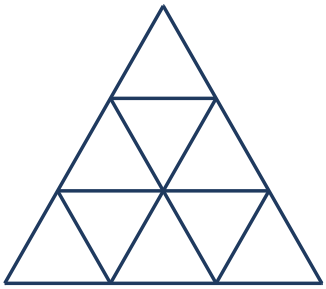
With k apex-parts and m lines parallel to the base inside the triangle, the count is **(m + 1) × k (k + 1) ÷ 2** (the +1 counts the base itself as one of the horizontal levels).



Example: 2 inner apex lines (k = 3) and 3 parallel lines. (3 + 1) × (3 × 4 ÷ 2) = 4 × 6 = **24**. (Program count: 24.)

3.3 Triangular grid (side divided into n equal parts)

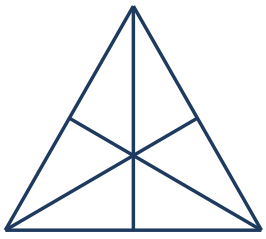
n	1	2	3	4	5	6
Upward triangles	1	4	10	20	35	56
Downward triangles	0	1	3	7	13	22
Total	1	5	13	27	48	78



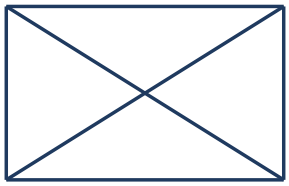
Total = ⌊ n (n + 2)(2n + 1) ÷ 8 ⌋. For n = 3: 3 × 5 × 7 ÷ 8 = 13.1, so **13**. By size for n = 3: 9 of size 1 (6 up + 3 down), 3 of size 2, 1 of size 3. 9 + 3 + 1 = 13 ✓. The up-pointing count is n (n + 1)(n + 2) ÷ 6.

3.4 Lines from vertices, diagonals and medians

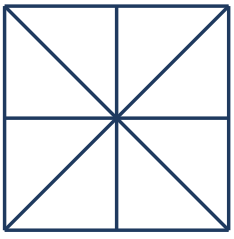
Figure	Triangles	Why
Rectangle with both diagonals	8	4 small triangles + 4 made of two of them (2 along each side pair)
Triangle with its 3 medians	16	6 small + others counted by size (verified by program)
Square with both diagonals and both midlines	16	8 small + 8 larger (verified by program)



Medians: 16



Rectangle: 8



Square: 16

SHORTCUT: For irregular figures, count by size: first all the smallest triangles, then those made of 2 regions, 3 regions, and so on. Tick or colour each group to avoid repeats.

CHAPTER 4

Counting Squares and Rectangles

Grids, tilted squares and the PYQ

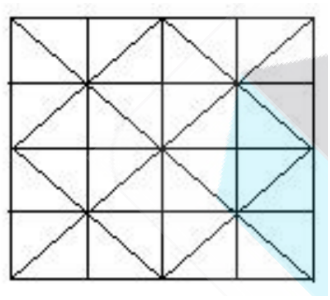
4.1 Grid of squares

Grid n × n	1	2	3	4	5	6
Squares n(n+1)(2n+1) ÷ 6	1	5	14	30	55	91
Rectangles (n(n+1) ÷ 2)²	1	9	36	100	225	441

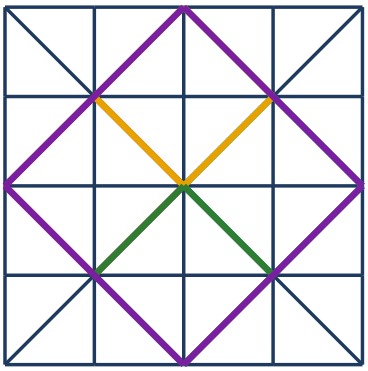
For an **m × n** grid: rectangles = [m (m+1) ÷ 2] × [n (n+1) ÷ 2]; squares = (m × n) + ((m − 1)(n − 1)) + ((m − 2)(n − 2)) + … until a factor is zero. Example 3 × 4: rectangles = 6 × 10 = 60; squares = 12 + 6 + 2 = 20.

4.2 English 2022 Q32: squares in a grid with diagonals

How many squares are there in the following figure? (A. 30 B. 35 C. 28 D. 33)



Original figure (from the paper)



Redrawn: the 5 tilted squares in colour

1. The figure is a 4 × 4 grid. Each of its 16 cells has one diagonal, drawn so that the diagonals form **X shapes at the points where they meet** (at the centre and at four other inner points).
2. **Squares with horizontal and vertical sides:** 16 (1 × 1) + 9 (2 × 2) + 4 (3 × 3) + 1 (4 × 4) = **30**.
3. **Tilted squares** (sides along the diagonals): the diagonals make 4 small diamonds around the centre (each has a diagonal of 2 cells) and 1 large diamond joining the midpoints of the four outer sides. That is 4 + 1 = **5**.
4. Total = 30 + 5 = 35. (A program counted all squares in the redrawn figure: 30 + 5 = 35.)
Answer: 35 (Option B)

TRAP: The common mistake is to stop at 30 (the plain grid). When diagonals meet at a point, look for **tilted squares** whose sides lie along the diagonals. Option A (30) is that trap.

CHAPTER 5

Shortcuts, Traps and Scope

Speed methods and mistakes

- **Apex lines:** k (k + 1) ÷ 2. **With parallel lines:** multiply by (m + 1).
- **Triangular grid:** remember 1, 5, 13, 27, 48, 78 for n = 1 to 6.
- **Square grid:** squares 1, 5, 14, 30, 55, 91; rectangles 1, 9, 36, 100, 225, 441.
- **Tilted squares:** after the plain count, check where diagonals cross and look for diamonds.
- **Count by size:** smallest first, then 2-region, 3-region and so on.

Trap	How to avoid it
Counting a triangle twice from two corners	Count each triangle once, by its three sides, or by its size and position.
Forgetting triangles that include the base	In apex figures, the base is one of the lines; include it.
Counting rectangles when asked for squares	Rectangles include squares; squares are only the equal-side ones.
Ignoring tilted squares	Look where diagonals meet (PYQ).
Treating three concurrent lines as a triangle	Three lines through one point form no triangle.

Scope

Related PYQs	Where
Eng 2022 Q40 (cutting a cube), Eng 2019 Q55, Eng 2022 Q53, Science 2024 Q96, SST 2023 Q13	Dice and cubes (done)
Eng 2019 Q47, Q48, Q49; Eng 2022 Q31	Paper Folding (coming)

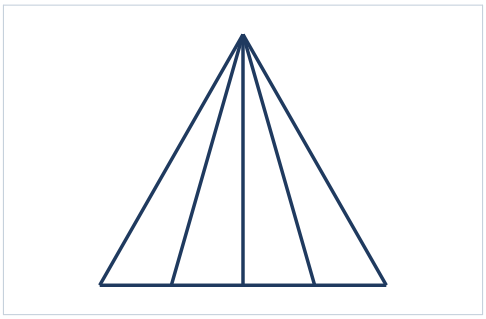
CHAPTER 6

Practice Set

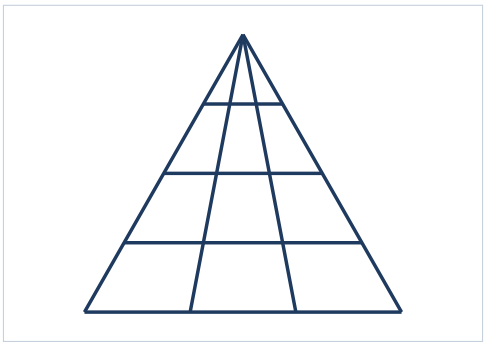
12 original counting questions; answers on the next page (all verified by program)

Part A: How many triangles are in the figure?

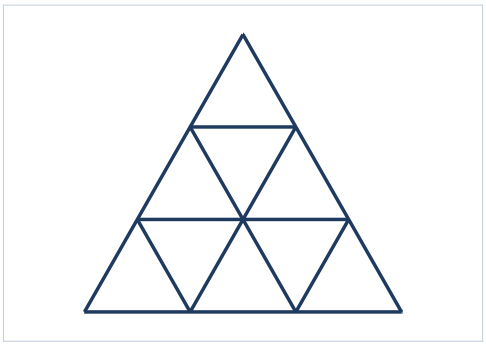
Q1. Count the triangles.



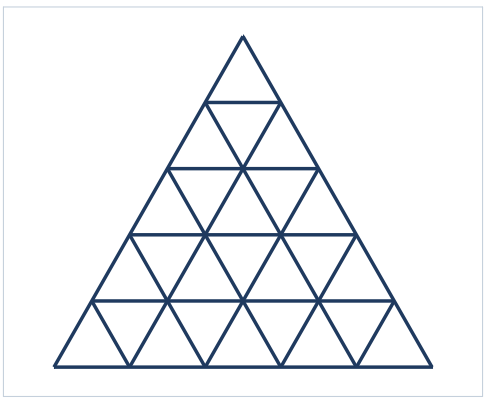
Q2. Count the triangles.



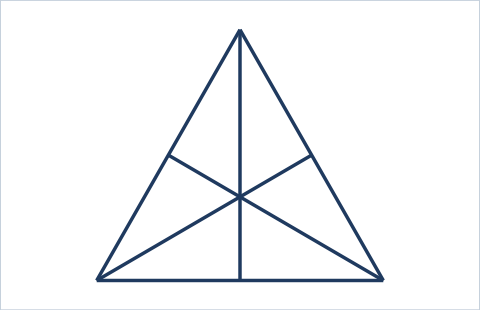
Q3. Count the triangles.



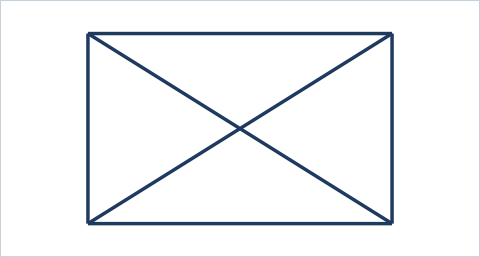
Q4. Count the triangles.



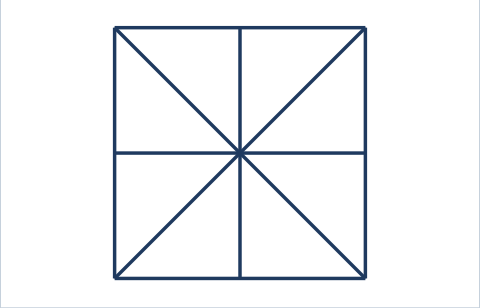
Q5. Count the triangles.



Q6. Count the triangles.

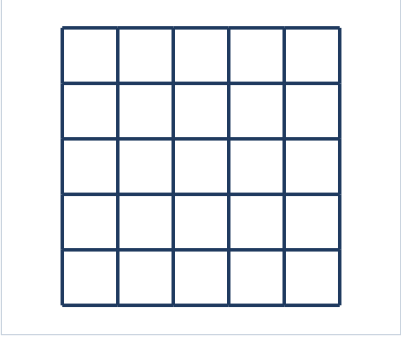


Q7. Count the triangles.

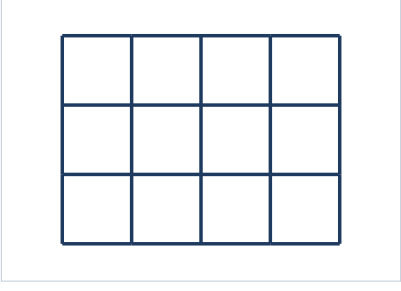


Part B: Squares and rectangles

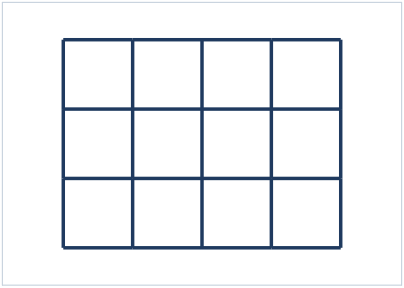
Q8. Count the squares in this 5 × 5 grid.



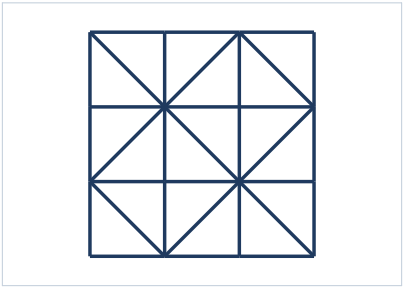
Q9. Count the rectangles (including squares) in this grid (4 columns, 3 rows).



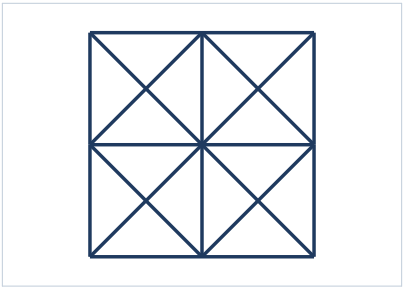
Q10. Count the squares in the same 4 × 3 grid.



Q11. Count all the squares in this 3 × 3 grid, which has one diagonal in every cell.



Q12. Count all the squares in this 2 × 2 grid with both diagonals drawn in every cell.



Answer key

Q	Answer	Working
1	10	k = 4: 4 × 5 ÷ 2
2	24	(m + 1) × k(k + 1)/2 = 4 × 6
3	13	triangular grid n = 3
4	48	triangular grid n = 5
5	16	three medians (program count)
6	8	rectangle with both diagonals
7	16	square with both diagonals and both midlines
8	55	$1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 1 + 4 + 9 + 16 + 25$
9	60	[4 × 5 ÷ 2] × [3 × 4 ÷ 2] = 10 × 6
10	20	12 + 6 + 2
11	16	14 axis-aligned (9 + 4 + 1) + 2 tilted = 16
12	6	5 axis-aligned (4 + 1) + 1 tilted = 6

CHAPTER 7

Quick Revision

Formula sheet

Figure	Formula or value
Apex lines (k parts on the base)	k (k + 1) ÷ 2: 1, 3, 6, 10, 15, 21
Apex lines + m parallel lines	(m + 1) × k (k + 1) ÷ 2
Triangular grid, side n	⌊ n (n + 2)(2n + 1) ÷ 8 ⌋: 1, 5, 13, 27, 48, 78
Rectangle with both diagonals	8
Triangle with 3 medians	16
Square with both diagonals + both midlines	16
n × n grid: squares	n (n + 1)(2n + 1) ÷ 6: 1, 5, 14, 30, 55, 91
n × n grid: rectangles	[n (n + 1) ÷ 2]²: 1, 9, 36, 100, 225, 441
m × n grid: rectangles	[m (m + 1) ÷ 2] × [n (n + 1) ÷ 2]
Grid with diagonals (PYQ figure)	30 plain + 5 tilted = 35

ROUTINE: **Exam-day routine:** identify the figure type, apply the formula, then count tilted squares if diagonals meet. 40 seconds per question. Next in the syllabus order: Rule Detection, Images, Completion of Pattern, Mirror Images, Figure Matrix, Paper Folding.